

Introduction to marketing mix modelling

International Marketing Analytics

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at
Office hours: book online (Google Calendar)

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SESSION 1 · WT 2026/27

Where should the next euro go?

Your situation

Many markets: different prices, media costs and cultures

Many channels: TV, online video, search, social, out-of-home

One budget: fixed, and negotiated every year

1. **Contribution:** what did each channel add to sales?
2. **Return:** which euro earned more, TV in France or paid social in Poland?
3. **Allocation:** how should next year's fixed budget be split?

The CEO's brief: next year's budget stays flat. Move money only where the next euro earns more.

What marketing mix modelling is

Part 1 of 7 · definition, inputs and outputs, MMM versus attribution and experiments, history

Marketing mix modelling in one sentence

MMM explains weekly sales with marketing and non-marketing drivers, splits sales into a base and the contribution of each activity, and turns these into budget decisions.

Aggregate data

Weekly totals per country and channel

No tracking of individuals, so privacy-safe

A regression model

Sales explained by spend, price, season and more

Plus carryover and diminishing returns

Decisions

What each euro returned

Where the next euro should go

Little (1979): a model must capture

- carryover
- diminishing returns
- competition
- interaction
- dynamics

Sources: Little 1979; Hanssens, Parsons & Schultz 2001; Hanssens & Pauwels 2016

Inputs, model, outputs

Inputs

- **Sales:** units or revenue per week
- **Media:** spend by channel
- **Marketing:** price, promotion, distribution
- **Season:** trend, holidays, weather
- **Context:** economy, competitors

Model

regression with carryover and saturation

Outputs

- **Contributions:** sales per driver
- **ROAS:** revenue per euro spent
- **Marginal ROAS:** revenue from the next euro
- **Response curves:** sales at each spend level
- **Allocation:** the best budget split

In practice: one row per week and country, from finance, media agencies and market research.

Three ways to measure marketing

	Marketing mix modelling	Multi-touch attribution	Experiments
Data	weekly totals per channel and country	clicks and views of individual users	test and control groups or regions
Answers	what each channel added, offline too	which touchpoints came before a sale	the causal lift of one change
Strength	all channels; privacy-safe	detailed and fast	most credible evidence of cause
Limit	correlation; needs variation in spend	digital only; credit is not cause	one channel, one period; costly

Triangulate: experiments calibrate the MMM, the MMM covers every channel, attribution steers digital campaigns day to day.

Sources: Google Modern Measurement Playbook 2023; IAB & MMA 2019; Gordon et al. 2019

An old method, popular again



1976

Clarke: how long ad effects last

1979

Little: what a response model needs

2001

Hanssens et al.: Market Response Models

2017

Jin et al. (Google): Bayesian MMM

2020s

Open source: Robyn, PyMC-Marketing, Meridian

Why now

- GDPR, cookie loss and Apple's tracking rules weaken user-level tracking (Part 2)
- MMM needs no personal data

64 % of senior marketing leaders use MMM (Gartner 2024)

13 % of advertisers turn it into insight quickly (WFA 2026)

Sources: Clarke 1976; Little 1979; Hanssens et al. 2001; Jin et al. 2017; Gartner 2024; WFA & Ebiquity 2026

Why attribution is not enough

Part 2 of 7 · where the money goes, last click, why attribution misleads, real-world tests

Where the advertising money goes

$1.14 tn global advertising spend in 2025, **+8.8 %**

73 % of it is digital: search, social, video, retail media

€119 bn digital advertising in Europe in 2024, **+16 %**

67 % of all ad spend in Europe is digital

What this means

- Digital ads are bought click by click
- The platforms that sell the ads also report their success
- Small measurement errors move billions

The question of this part: who gets the credit for a sale, and is it deserved?

Sources: WPP Media, This Year Next Year, mid-year 2025; IAB Europe, AdEx Benchmark 2024

Attribution: who gets the credit?



Credit for the order	TV	Instagram	YouTube	Search
Last click	**0 %**	0 %	0 %	100 %
First click	**0 %**	100 %	0 %	0 %
Linear	**0 %**	33 %	33 %	33 %
Data-driven (GA4)	**0 %**	e.g. 25 %	e.g. 15 %	e.g. 60 %

All four models share out one sale: none asks if it would have happened without ads.

Sources: Google Analytics 4 documentation; journey and data-driven shares are illustrative

Last click: simple, common and biased

How it works

- 100 % of the credit to the last ad clicked before the order
- Easy to compute, easy to explain
- GA4 and Google Ads: since 2023 only last click and data-driven are left

Who wins, who loses

- Wins: branded search, retargeting, coupon and affiliate sites
- Loses: TV, online video, social, posters
- Ignores: views without a click, offline sales

The catch: the last click often comes from people who had already decided to buy

Example: after a TV spot, a viewer searches the brand name: search gets the sale, TV gets nothing

The result: budgets drift to the end of the funnel, and the demand that feeds it slowly dries up

Sources: Search Engine Land 2023 (Google Ads and GA4); Li & Kannan 2014 (Journal of Marketing Research)

Six problems of attribution models

Correlation only

Ads target people who buy anyway

Credit is not extra sales

Tracking gaps

Cookie consent, ad blockers, Apple's ATT

2021: 4 % of US iPhone users allowed tracking

Offline is invisible

TV, radio, posters, word of mouth

Sales in shops have no click

Double counting

Each platform counts its own orders

Sum of reports > real orders

Click fraud

Affiliates claim sales that come anyway

Uber: fake clicks on free installs

Many devices

Phone on the sofa, laptop at work

One buyer looks like three

The remedy: experiments (ads off in some regions) and MMM, which needs no tracking.

Sources: Gordon et al. 2019 (Marketing Science); Flurry Analytics 2021; WARC 2019 (Uber)

When the ads were switched off

99.5 %	**eBay:** brand-keyword traffic that came back through the free links when the ads were paused
$100 m	**Uber:** of $150 m yearly performance ads switched off, with no change in app installs
$200 m	**P&G:** of digital ads cut in 2017; reach rose about 10 %
95 %	**Airbnb:** of web traffic kept in 2020 after cutting marketing by more than half
15	**Facebook experiments:** click-based methods often missed the true lift, mostly overstating it

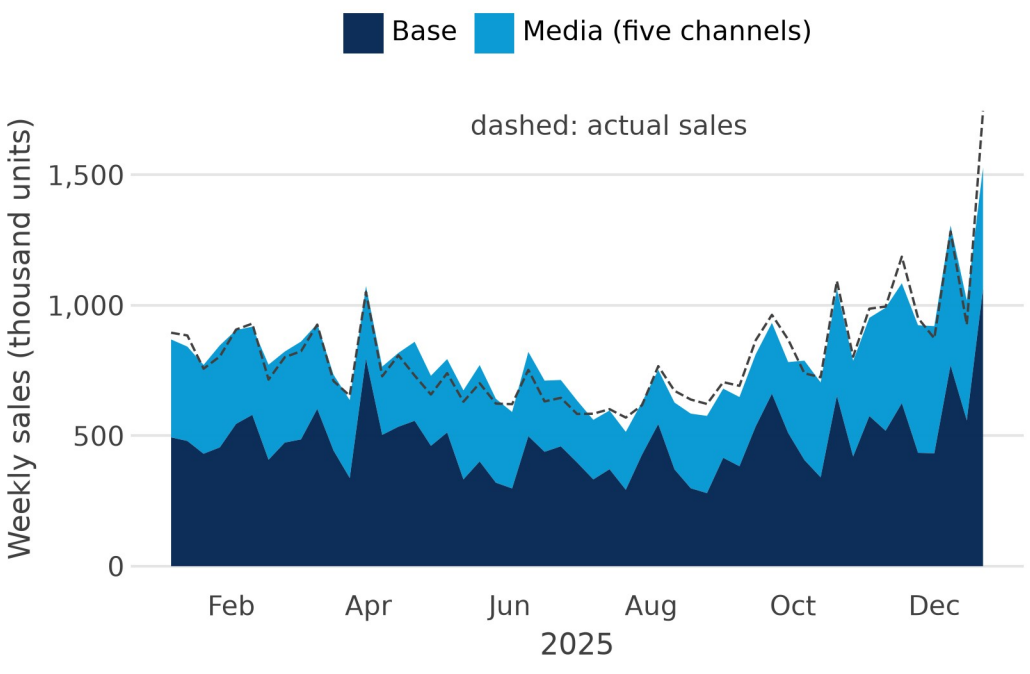
Lesson: test what the ads add; do not trust the click report.

Sources: Blake, Nosko & Tadelis 2015; WARC 2019; Adweek 2018; Campaign Asia 2021; Gordon et al. 2019

How MMM works

Part 3 of 7 · base and incremental sales, the equation, adstock, saturation, seasonality

Sales = base + incremental



Base: what would sell without media: price, distribution, season, trend

Incremental: what media add on top, channel by channel

Contribution: the sales that would disappear if a channel were switched off

Naive model: media ≈ 41 %, too high

Simulated data, Germany, 2025, naive regression: budgets follow Christmas demand (see the pitfalls).

Sources: Hanssens et al. 2001; Jin et al. 2017; own calculation on simulated data

The regression behind it

Spend per channel → **Adstock: carryover** → **Saturation: less per €** → **× β: the effect** → **Sales contribution**

Additive form

```
sales_t = β0 + Σ βc·f(spend_c,t)
          + γ·controls_t + ε_t
```

- **βc:** extra units per unit of transformed spend
- **Good for:** contributions in units or euros

Log-log form

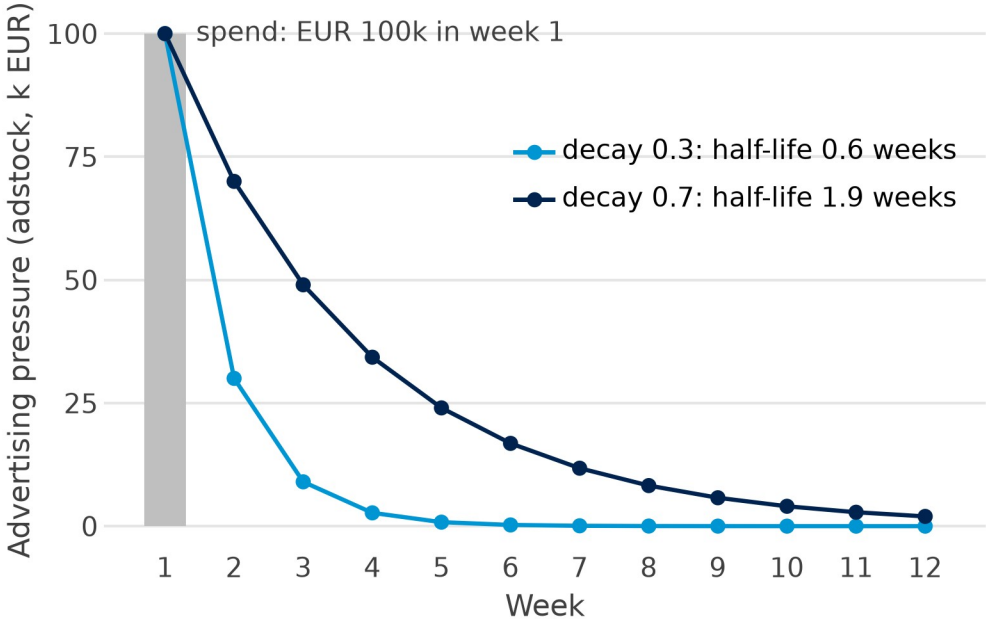
```
ln sales_t = β0 + βp·ln price_t
   + Σ βc·ln(1 + adstock_c,t) + … + ε_t
```

- **β:** elasticity, % change in sales per 1 % change
- **Good for:** comparing countries and drivers

Controls: price, promotion, distribution, trend, season, holidays, weather, economy, competitor spend.

Sources: Hanssens et al. 2001; Tellis 2006; Jin et al. 2017 · Refresher: deck 01b, Parts 6 and 9

Adstock: advertising keeps working



Geometric adstock

```
A_t = spend_t + λ·A_(t−1)
```

λ (decay): share of last week's pressure that carries over

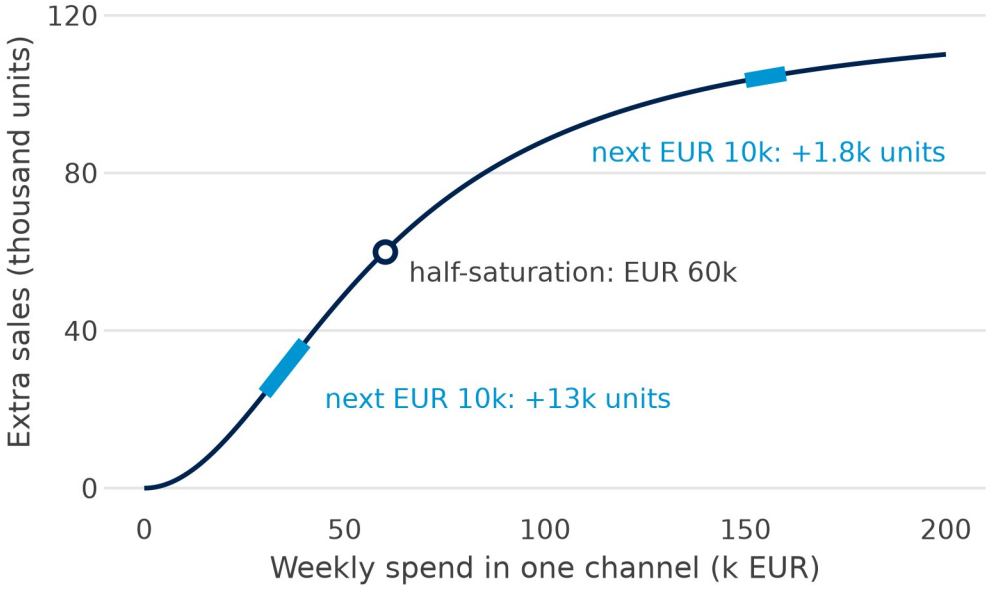
Half-life: ln 0.5 / ln λ = 0.6 or 1.9 weeks here

Total effect: 1/(1−λ) × week 1: 1.4× versus 3.3×

Without adstock, the weeks after a campaign are credited to whatever else happened then.

Sources: Little 1979; Jin et al. 2017; Robyn documentation (geometric and Weibull adstock)

Saturation: each extra euro buys less



Hill saturation

```
Hill(x) = x^s / (x^s + K^s)
```

K: spend at half the maximum effect

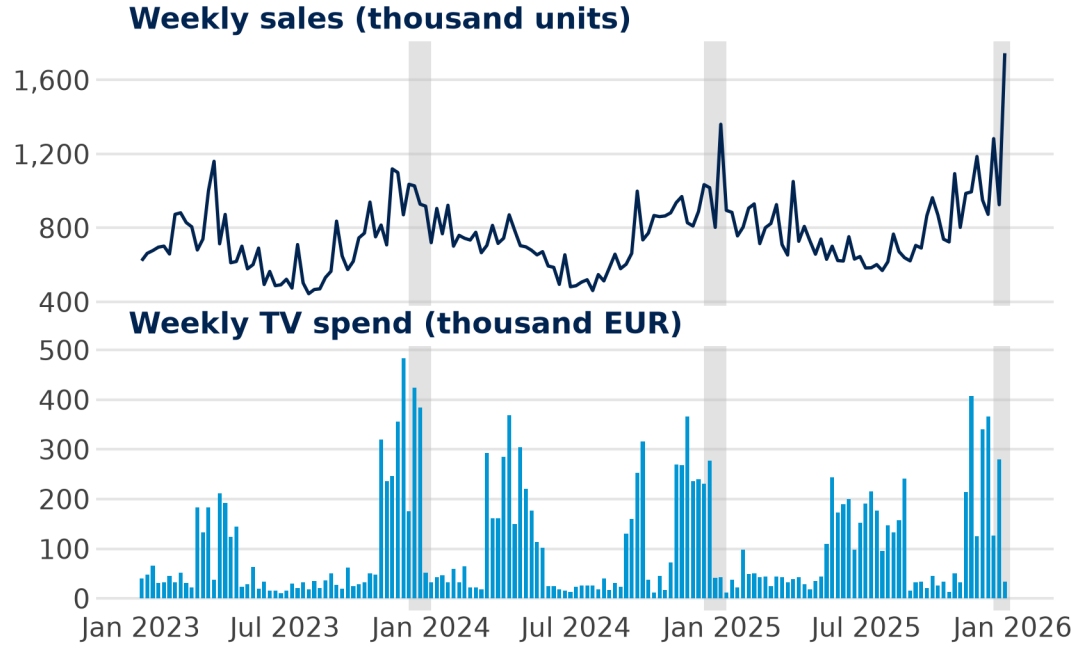
s: shape; above 1 gives an S-curve

Marginal ROAS: revenue from the next euro, the slope of the curve

Decide at the margin: move money to where the next euro earns most.

Sources: Jin et al. 2017; Google Meridian documentation (media saturation and lagging)

Seasonality, trend and holidays



Simulated data, Germany, 2023–2025; grey: Christmas weeks.

Trend: slow growth or decline, a week counter

Seasonality: the yearly wave, Fourier terms

Holidays: dummies for Christmas, Easter, Valentine's

Weather, economy: temperature, consumer confidence

TV peaks when sales peak: does TV drive Christmas, or Christmas the TV budget?

Key terms (1): what the model splits

Marketing mix

The levers a brand controls: product, price, place (distribution) and promotion.

Example: price, promotions, shelf listings and five media channels

Base sales

What would sell without media: driven by price, distribution, season and trend.

The dark area in the decomposition chart

Incremental sales

What a channel adds on top of the base: the sales lost if it were switched off.

Also called the channel's contribution

Adstock

Advertising keeps working after the week it runs; the decay is the share that carries over.

Decay 0.7: half the effect is gone after about 2 weeks

Saturation

Each extra euro buys less: the response flattens as spend grows.

The next EUR 10k buys far fewer units at high spend

Control variables

Non-media drivers in the model, so media are not credited with their effect.

Price, promotions, holidays, temperature, competitors

Key terms (2): what the model tells you

Elasticity

The % change in sales for a 1 % change in a driver; the coefficient in a log-log model.

−2: a 10 % lower price sells 20 % more units (HBR reading)

Response curve

Incremental sales at each spend level: adstock and saturation in one picture.

Read it to see where a channel stops paying off

ROAS and ROI

ROAS: incremental revenue per euro spent. ROI: incremental profit per euro spent.

ROAS 3: each euro returned EUR 3 of revenue

Marginal ROAS

The return on the next euro: the slope of the response curve.

Move budget to the highest marginal, not average, ROAS

Holdout

Fit on past weeks, predict weeks the model has not seen, compare the errors.

Fit two years, forecast the third, report the MAPE

Calibration

Checking or anchoring model effects with experiments such as geo lift tests.

Geo test: more spend in some regions, compare with the rest

What we know: empirical generalisations

Part 4 of 7 · elasticities, carryover and wear-out, differences between countries

Elasticities: price versus advertising

Advertising: the meta-analyses

0.12 short-term elasticity, mean (Sethuraman et al. 2011)

0.24 long-term elasticity, mean (same study)

0.09 mean in an international database (Henningsen et al. 2011)

≈ ½ of 389 TV weight tests: no significant sales effect (Lodish et al. 1995)

Price, for comparison

−2.62 mean price elasticity, 1,851 estimates (Bijmolt et al. 2005)

−2 Event Network: 10 % lower price, 20 % more units (HBR reading)

Per per cent, price moves sales about 20 times more than advertising.

Reading 0.12: 10 % more advertising lifts sales by about 1.2 %.

Carryover and wear-out

How long effects last

60 % of the effect came after week 13 (141 UK brands; Ebiquity & Thinkbox 2024)

2× long-term elasticity versus short-term: 0.24 against 0.12 (Sethuraman et al. 2011)

Data interval matters

Monthly or annual data suggest longer carryover than weekly data (Clarke 1976; Köhler et al. 2017)

Wear-in and wear-out

Ads may need repetition to work, and lose effect when shown too often or too long (Tellis 2004; Vakratsas & Ambler 1999)

For your model: estimate carryover per channel from weekly data and check half-lives against these benchmarks.

Effects differ between countries

Market type: advertising matters more in emerging markets; price and distribution effects differ too (Bahadir et al. 2015, 14 markets)

Business cycle: in downturns price sensitivity rises and advertising works less (van Heerde et al. 2013)

Culture: how strongly ad budgets follow the economy depends on national culture (Deleersnyder et al. 2009, 37 countries)

Payoff: reallocating across countries with elasticities: about EUR 500m more profit at Bayer (Fischer et al. 2011)

For international brands: expect markets to respond differently. Hierarchical models let countries differ while borrowing strength from each other (Sun et al. 2017).

Doing it well

Part 5 of 7 · data, pitfalls, validation, Bayesian priors, open-source tools

What the data must offer

Two to three years, weekly

Seasons must repeat: three years are 156 weeks per country

Variation in spend

Channels must move up and down, and not always together

Fine geography

Countries or regions, not one national total (Nielsen 2022)

All drivers

Price, distribution, season and competitors, not only media

+68 % **A media-only model** overstated advertising ROI by 68 % compared with a model that included the other drivers (Nielsen 2022).

Four pitfalls

Endogeneity

Budgets follow demand: TV peaks before Christmas, when people buy anyway

Multicollinearity

Channels that move together cannot be told apart; check the VIF

Overfitting

Many parameters, few weeks: the fit looks great, the forecast fails

Identification

Saturation and effects that change over time can mimic each other

10.6 vs 4.2 **Paid-search ROAS:** a standard MMM against geo experiments at Zalando (Heusch 2026).

Sources: Chan & Perry 2017; Dew et al. 2024; Heusch 2026 · VIF, tests, endogeneity: deck 01b, Part 7

Validate before you trust

Fit → Holdout forecast → Plausibility checks → **Calibrate with experiments**

Holdout

- Fit to two years
- Forecast the third
- Compare errors (MAPE)

Plausibility

- Media effects positive
- Half-lives and elasticities near benchmarks
- Stable when data change

Experiments

- Spend up or off in test regions
- Lift = test minus control
- Geo tests: test versus control regions

Why experiments: across 663 Facebook experiments, non-experimental estimates were often far from the true lift (Part 2 showed the switch-off tests).

Sources: Vaver & Koehler 2011; Gordon et al. 2019; Gordon, Moakler & Zettelmeyer 2023

Bayesian MMM: start from what we know

Prior

what we believe first: elasticities near 0.1, a lift-test result

+

Data

156 weeks of sales, spend and controls per country

→

Posterior

updated belief: every effect and ROAS with a range

Why it suits MMM

- Short series, many parameters: priors keep estimates sensible
- Uncertainty for every output
- Experiments enter as priors

What to watch

- Weak data: the prior decides; compare prior and posterior
- Check that the sampler converged
- State and justify every prior

Sources: Jin et al. 2017; Hanssens 2018; Zhang et al. 2024 (Google)

Open-source MMM tools

	Robyn	Meridian	PyMC-Marketing
Maker	Meta	Google	PyMC Labs
Language	R (Python beta)	Python	Python
Method	ridge regression with automated search	hierarchical Bayesian, by geo	Bayesian, modular
Strengths	calibration, budget allocator, widely used	many geos, reach and frequency, ROI priors	lift-test calibration, time-varying effects, optimiser
Version	3.12.0 (Dec 2024)	2.1.0 (Sep 2026)	1.2.0 (Sep 2026)

In this course: PyMC-Marketing and Meridian in Python, Session 5; set up your AI assistant with deck 01d.

Sources: Runge et al. 2024; Meridian and PyMC-Marketing documentation; open-source MMM review (2026)

Reading: Analytics for Marketers

Part 6 of 7 · Fantini & Narayandas (HBR 2023)
and an in-class discussion

Three approaches to analytics

Descriptive: humans decide → **Predictive: machines predict, humans choose** → **Prescriptive: machines decide**

What happened?

- Dashboards of aggregated past data
- Managers add experience and gut feeling
- Low pain, low gain

What if ...?

- Regression, elasticities, forecasts
- Finer, tactical decisions
- Few variables, uncertain inputs

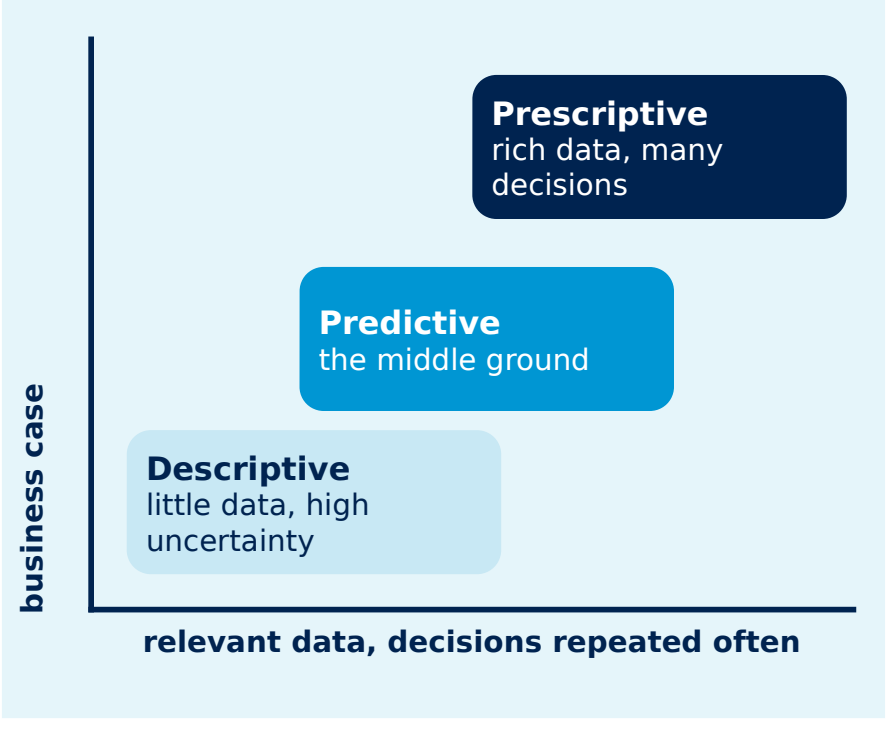
What to do?

- Optimises towards goals and rules set by managers
- Learns from many cheap experiments
- Costly to build and run

Keep in mind: where does MMM sit? You decide in the discussion.

Sources: Fantini & Narayandas 2023 (Harvard Business Review)

When to use which approach



Humans lead: strategy, innovation, entering a new market

Predictive fits: planning, noisy customer data, small gains from fine-tuning

Prescriptive pays: prices, stock, marketing spend, decided often with rich data

Sources: Fantini & Narayandas 2023 (Harvard Business Review); simplified illustration

Case: markdowns at Event Network

1 Descriptive

- Discount SKUs with the most days of stock
- Until the markdown budget is spent
- Ignores customers and context

2 Predictive

- Regression of volume on price
- Elasticity by category, store and week
- Crude, but better than the rule

3 Prescriptive

- Machine learning plus optimisation
- Many data sources, 100,000+ SKUs
- Learns and improves over time

−2
price elasticity

+20 %
units after a 10 % price cut

+8 %
revenue: $1,000 → $1,080

10–20 %
of sales explained by price

Lesson: a crude model can still beat a rule of thumb.

Sources: Fantini & Narayandas 2023 (Harvard Business Review)

Accuracy is not the goal

Data scientists

- Aim: the most accurate model
- Judge by statistical fit

Business scientists

- Aim: the most profitable decision
- Judge by the cost of each error

False positive: predicted a win that does not come: wasted sales and marketing effort

False negative: predicted a loss that would have been a win: a lost opportunity

In this course: Session 3 and case task C7 set the churn threshold by profit, not accuracy

The manager's new role: “from the person who has all the answers to the one who asks the right questions”

Sources: Fantini & Narayandas 2023 (Harvard Business Review)

In-class discussion (1): classify the decisions

	Decision of an international brand manager	How often	Data available
1	Split the global media budget over 20 countries and 6 channels	once a year	3 years of weekly sales and spend per country
2	Set the bids for search keywords in each market	every day	clicks, costs and sales per keyword
3	Set the launch price in a newly entered market	once per market	almost none from that market
4	Decide whether to enter Brazil	once	market studies, no own sales
5	Mark down 10,000 SKUs in 300 stores after the season	every season	sales and stock per SKU and store
6	Shift digital budget between channels by last-click reports	every month	clicks of tracked users only

Your task: classify each decision as descriptive, predictive or prescriptive, using the article's two criteria, and say who should decide.

In-class discussion (2): MMM for international brands

Groups of 3–4: 15 minutes → **Plenary: 10 minutes** → **Hand in: one photo of your answers**

1. **MMM in the framework:** descriptive, predictive or prescriptive? What would make it prescriptive?
2. **International brand managers:** how can MMM help them? Name two decisions across countries it improves.
3. **Crude models:** Event Network's price model explained 10–20 % of sales. Is a rough MMM for a small market still worth it?
4. **Errors:** the MMM says TV does not pay in one country. What would a false negative cost?
5. **The manager's job:** name one international decision no model should make, with the article's argument.

Sources: Reading: Fantini & Narayandas 2023, Analytics for Marketers, Harvard Business Review (on Canvas)

MMM in this course

Part 7 of 7 · how the sessions build the skills,
the group case study

Five sessions, one model

1	**Introduction and linear regression**	OLS, model fit and interpretation	**→ the base regression**
2	**Advanced regression**	log-log elasticities, dummies, non-linear effects, moderation and mediation, adstock and saturation, diagnostics	**→ the MMM itself**
3	**Logistic regression**	binary outcomes: purchase, churn; odds ratios	**→ customer-level questions**
4	**ARIMA**	trend, seasonality, ARIMA and ARIMAX forecasts	**→ the sales baseline**
5	**Advanced MMM tools**	Bayesian MMM, budget optimisation	**→ the full workflow**

Where to find what

In this deck	Where to look
Regression, log-log, elasticities	deck 01b, Parts 6 and 9; sessions/01-foundations/regression_chocolate.qmd
VIF, tests, endogeneity (pitfalls)	deck 01b, Part 7: assumptions and tests
Coding in Python with an AI assistant	deck 01d: GitHub Copilot, Claude Code, Codex in Positron
Practice: regression on real data	coding exercise 1: your personal brief and data on Canvas
The full MMM, step by step	Sessions 2 to 5 and the group case study



All decks and class files: the course repository; pull it before every session.

The group case study

S1: what drives weekly orders? → **S2: price and advertising** → **S3: who will cancel?** → **S4: Q1 2026 baseline**

The brief

- Alpenglow's web shop in AT, DE and CH
- Four questions from the Head of E-commerce

The data

- Weekly shop data, 3 countries × 156 weeks
- Club subscribers
- The Q1 2026 plan

You deliver

- A Quarto report with all code
- Groups of at most 4
- 30 % of the grade

Every number from code: the report renders from a fresh copy of your project folder.

Brief, data and assessment criteria: `assignments/group-project/` and Canvas

Key takeaways

1. **MMM** explains weekly sales with marketing and other drivers and splits them into base and incremental

2. **Adstock and saturation** capture carryover and diminishing returns; decide on marginal, not average, ROAS

3. **Benchmarks:** advertising elasticities around 0.1 short term, larger price effects, differences by country

4. **Clicks are not causes:** attribution shares out sales; validate MMM on a holdout and calibrate it with experiments

5. **Open-source Bayesian tools** make it doable in Python: you will build one



The question stays the same: where should the next euro go?

Further reading

Foundations and evidence

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Sethuraman, R., Tellis, G. J. & Briesch, R. A. (2011). How well does advertising work? Journal of Marketing Research 48(3), 457–471.

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VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

INSTITUTE FOR INTERNATIONAL BUSINESS
Welthandelsplatz 1, 1020 Vienna, Austria

DR ARNE FLOH
Senior Lecturer in Marketing

F +43-31336-6367
arne.floh@wu.ac.at
www.wu.ac.at/en/welthandel/arne-floh
Office hours: book online